

# Structural Calculation Report

Reinforced Concrete Beam Design to BS EN 1992-1-1:2004+A1:2014 (Eurocode 2)

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## Executive Summary

|                 |                                       |                  |                            |
|-----------------|---------------------------------------|------------------|----------------------------|
| Overall Status: | <b>FAIL</b>                           | Max Utilisation: | $\eta = 123.5\%$           |
| Section:        | 300.0×500.0 mm                        | Span:            | 6000.0 mm                  |
| Main Steel:     | 3-Ø16<br>(As=603.19 mm <sup>2</sup> ) | Links:           | Ø8 @ 340.5 mm c/c (2 legs) |

**Recommendation:** Provided steel insufficient. Use 1-Ø32 (As=804.2 mm<sup>2</sup>,  $\eta=80\%$ ).

## 1. Design Parameters

|                                          |                                 |     |
|------------------------------------------|---------------------------------|-----|
| Section Type                             | Rectangular                     |     |
| Web width ( $b_w$ )                      | 300.0                           | mm  |
| Overall depth (h)                        | 500.0                           | mm  |
| Nominal cover ( $c_{nom}$ )              | 30.0                            | mm  |
| Span (L)                                 | 6000.0                          | mm  |
| Concrete grade                           | C30.0/37 ( $f_{ck} = 30.0$ MPa) |     |
| Steel yield strength ( $f_{yk}$ )        | 500.0                           | MPa |
| Partial factor — concrete ( $\gamma_c$ ) | 1.5                             |     |
| Partial factor — steel ( $\gamma_s$ )    | 1.15                            |     |
| $\alpha_{cc}$                            | 0.85                            |     |

## Applied Actions

|                                    |       |     |
|------------------------------------|-------|-----|
| Design bending moment ( $M_{Ed}$ ) | 120.0 | kNm |
| Design shear force ( $V_{Ed}$ )    | 80.0  | kN  |

## 2. Bending Design EN 1992-1-1 §6.1

Effective depth:

$$d = h - c_{nom} - \phi_{link} - \phi_{bar}/2$$

$$d = 500.0 - 30.0 - 8.0 - 16.0/2$$

$$\rightarrow 454.0 \text{ mm}$$

Design concrete strength:

$$f_{cd} = \alpha_{cc} \cdot f_{ck} / \gamma_c$$

$$f_{cd} = 0.85 \times 30.0 / 1.5$$

$$\rightarrow 17.00 \text{ MPa}$$

Design steel strength:

$$f_{yd} = f_{yk} / \gamma_s$$

$$f_{yd} = 500.0 / 1.15$$

$$\rightarrow 434.78 \text{ MPa}$$

K-factor:

$$K = M_{Ed} / (b_{eff} \cdot d^2 \cdot f_{ck})$$

$$K = 120.0 \times 10^3 / (300.0 \times 454.0^2 \times 30.0)$$

$$\rightarrow 0.0647 \text{ (} K' = 0.167 \text{)}$$

Lever arm:

$$z = d \cdot (0.5 + \sqrt{(0.25 - K/1.134)})$$

$$z = 454.0 \times (0.5 + \sqrt{(0.25 - 0.0647/1.134)})$$

$$\rightarrow 426.43 \text{ mm}$$

Required tension steel:

$$A_{s, req} = M_{Ed} / (f_{yd} \cdot z)$$

$$A_{s, req} = 120.0 \times 10^3 / (434.78 \times 426.43)$$

$$\rightarrow 647.24 \text{ mm}^2$$

Provided tension steel:

$$A_{s, prov} = n \cdot \pi \cdot \phi^2 / 4$$

$$A_{s, prov} = 3 \times \pi \times 16.0^2 / 4$$

$$\rightarrow 603.19 \text{ mm}^2$$

|                                        |                          |                 |
|----------------------------------------|--------------------------|-----------------|
| Effective depth (d)                    | 454.0                    | mm              |
| K-factor                               | 0.0647                   | (K' = 0.167)    |
| Neutral axis ratio (x/d)               | 0.152                    |                 |
| Lever arm (z)                          | 426.43                   | mm              |
| Required A <sub>s</sub>                | 647.24                   | mm <sup>2</sup> |
| Minimum A <sub>s,min</sub>             | 205.14                   | mm <sup>2</sup> |
| Provided A <sub>s,prov</sub> (3-Ø16)   | 603.19                   | mm <sup>2</sup> |
| Moment capacity (M <sub>Rd</sub> )     | 111.83                   | kNm             |
| Utilisation ( $\eta = M_{Ed}/M_{Rd}$ ) | 107.3%                   |                 |
| Status                                 | FAIL: INSUFFICIENT_STEEL |                 |

### 3. Shear Design EN 1992-1-1 §6.2

Size factor:

$$k = \min(2.0, 1 + \sqrt{(200/d)})$$

$$k = 1 + \sqrt{(200/454.0)}$$

→ **1.664**

Concrete shear resistance:

$$V_{Rd,c} = [C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3}] \cdot b_w \cdot d$$

$$\rho_l = 0.00443$$

→ **64.40 kN**

Max strut capacity:

$$V_{Rd,max} = \alpha_{cw} \cdot b_w \cdot z \cdot v \cdot f_{cd} / (\cot\theta + \tan\theta)$$

$$\theta = 21.8^\circ, \cot\theta = 2.50$$

→ **446.36 kN**

Required shear reinforcement:

$$A_{sw/s} = V_{Ed} / (z \cdot f_{ywd} \cdot \cot\theta)$$

$$A_{sw/s} = 80.0 \times 10^3 / (408.6 \times 434.8 \times 2.50)$$

→ **0.263 mm<sup>2</sup>/mm**

|                                  |               |                        |
|----------------------------------|---------------|------------------------|
| $V_{Ed}$                         | <b>80.0</b>   | kN                     |
| $V_{Rd,c}$ (concrete resistance) | <b>64.4</b>   | kN                     |
| $V_{Rd,max}$ (strut capacity)    | <b>446.36</b> | kN                     |
| Strut angle ( $\theta$ )         | <b>21.8°</b>  | ( $\cot\theta = 2.5$ ) |
| Required $A_{sw}/s$              | <b>0.263</b>  | mm <sup>2</sup> /mm    |
| Max link spacing ( $s_{max}$ )   | <b>340.5</b>  | mm                     |
| Provided: Ø8 @ 340.5 mm          | <b>2 legs</b> |                        |
| Utilisation ( $\eta$ )           | <b>61.0%</b>  |                        |
| Status                           | <b>OK</b>     |                        |

#### 4. Deflection Check EN 1992-1-1 §7.4.2

Reference reinforcement ratio:

$$\rho_{\min} = \sqrt[3]{f_{ck}} \times 10^{-3}$$

$$\rho_{\min} = \sqrt[3]{30.0} \times 10^{-3}$$

→ **0.00548**

Actual reinforcement ratio:

$$\rho = A_{s,req} / (b \cdot d)$$

$$\rho = 647.24 / (300.0 \times 454.0)$$

→ **0.00475**

Basic span/depth ratio:

$$L/d_{basic} = K \cdot [11 + 1.5\sqrt{f_{ck}} \cdot \rho_{\min} / \rho + \dots]$$

$$K_{sys} = 1.0$$

→ **21.51**

Modification factor:

$$\text{Factor} = (500/f_{yk}) \cdot (A_{s,prov}/A_{s,req}) \leq 1.5$$

$$\text{Factor} = (500/500.0) \times (603.2/647.2)$$

→ **0.932**

Allowable span/depth:

$$L/d_{allow} = L/d_{basic} \times \text{Factor}$$

$$21.51 \times 0.932$$

→ **20.05**

Actual span/depth:

$$L/d_{actual} = L / d$$

$$6000.0 / 454.0$$

→ **13.22**

|                 |       |  |
|-----------------|-------|--|
| Actual L/d      | 13.22 |  |
| Allowable L/d   | 20.05 |  |
| Utilisation (η) | 65.9% |  |
| Status          | OK    |  |

#### 5. Cracking Control EN 1992-1-1 §7.3.3

Estimated steel stress:

$$\sigma_s = 310 \cdot (f_{yk}/500) \cdot (A_{s,req}/A_{s,prov})$$

$$\sigma_s = 310 \times (500.0/500) \times (647.2/603.2)$$

→ **332.6 MPa**

Max allowable spacing:

From Table 7.3N for  $w_k = 0.3$  mm

$$\sigma_s = 332.6 \text{ MPa} \rightarrow \text{interpolate}$$

→ **84.2 mm**

Actual bar spacing:

$$s = (b - 2c - 2\phi_{link} - n \cdot \phi_{bar}) / (n-1) + \phi_{bar}$$

$$n = 3, \phi = 16.0 \text{ mm}$$

→ **104.0 mm**

|                                       |        |     |
|---------------------------------------|--------|-----|
| Estimated steel stress ( $\sigma_s$ ) | 332.6  | MPa |
| Actual bar spacing                    | 104.0  | mm  |
| Max allowable spacing (Table 7.3N)    | 84.2   | mm  |
| Crack width limit ( $w_k$ )           | 0.3    | mm  |
| Utilisation ( $\eta$ )                | 123.5% |     |
| Status                                | FAIL   |     |

## 6. Utilisation Summary

| Check               | Utilisation ( $\eta$ ) | Status |
|---------------------|------------------------|--------|
| Bending (§6.1)      | 107.3%                 | ✗      |
| Shear (§6.2)        | 61.0%                  | ✓      |
| Deflection (§7.4)   | 65.9%                  | ✓      |
| Cracking (§7.3)     | 123.5%                 | ✗      |
| Maximum Utilisation | 123.5%                 | FAIL   |

Assumptions & Limitations

- Simply supported beam with uniform loading assumed for deflection check.

- Concrete properties derived from Table 3.1 of EN 1992-1-1.

- Exposure class: XC1 (unless otherwise specified). Minimum cover per Table 4.4N.

- Shear design uses Variable Strut Inclination Method per §6.2.3.

- Deflection checked using simplified span/effective depth method per §7.4.2 (Table 7.4N).

- Cracking control via bar spacing rules per §7.3.3 (Table 7.3N). Direct crack width calculation not performed.

- Quasi-permanent load ratio assumed as 0.6 for steel stress estimation in cracking check.

- Fire resistance, durability detailing, and anchorage/lap lengths are outside the scope of this calculation.

- This report is for preliminary design purposes. Independent checking is recommended for final design.